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CSE-AI

Short answer

1 A college wants to identify students who may fail in the final exam. How can Data Science help?

Data Science can analyze students marks, attendance, assignment scores, and previous that indicate which students are likely to perform poorly. This helps the college take early action and improve student performance.

. Teachers can use these results to identify students who need additional support.

2 A hospital has patient records for many years. How can Data Science be applied?

Data Science can analyze hertorical patient records to identify disease patterns, treatment outcomes, and health trends. It can help doctors make better decisions. It can also help hospitals improve planning and patient care.

3 An Online shopping website wants to recommend Products to customers. which use of Data Science is applied here?

This is an application of recommendation systems in Data Science. The System analyzes customers' Previous purchases, Searches and product preferences. It uses these patterns to suggest Products that customers may be interested in. This can improve the shopping experience and help increase Sales

4 A bank wants to detect suspicious transactions. How can Data Science be useful?

Data Science can analyze large numbers of bank transactions and identify unusual patterns. It can compare current transactions with normal customer transaction behaviour.

Transactions that show unusual amounts, locations, or activities can be flagged for investigation.

5 A company wants to reduce customer loss. How can Data Science support this?

Data Science can analyze customer purchases, usage patterns, complaints and feedback. It can identify customers whose behaviour indicates that they may stop using the company's services. The company can provide offers, better support or personalized services to such customers. This helps improve customer satisfaction and reduce customer loss.

6 A school wants to improve student performance, mention one benefit of Data Science.

Data Science can analyze students marks, attence, assignments, and learning patterns. It can help identify students who are weak in particular subjects.

Teachers can provide additional class, materials, or individual support to those students.

Thus, Data Science helps improve students academic performance

A supermarket wants to know which products sell more during festivals. How can Data Science Help?

Data Science can analyze previous festival sales and customer purchasing patterns. It can identify the products that have higher-demand during different festivales.

The supermarket can use this information to maintain sufficient stock of popular products.

This helps reduce shortage and improve sales during the festival period.

8 A transport department wants to reduce traffic congestion. How can Data Science be applied?

Data Science can analyze traffic, vehicle movement, and road data to identify congestion patterns and improve traffic management.

9 Farmer wants to improve crop yield. How DS be applied?

Data Science can analyze weather, soil, irrigation, and crop data to recommend suitable farming conditions and improve crop yield.

10 Excel, tables, reviews, images and videos data in a company. which facet of data is shown here?

This shows the variety facet of data because the data exists in different formats such as structured tables, text, images and videos.

Data science must handle these different forms of data for analysis.

11 A weather app updates temperature every minute. which facet of data is involved?

The facet involved is velocity. velocity refers to the speed at which data is generated, collected and updated.
In this case, the temperature information is continuously updated every minute.
Therfore, the weather application deals with data that has a high rate generation.

12 A university stores students details in rows and columns. what type of data structure is this?

The data is called structured data. Structured data is organized in a fixed using rows and columns. Student details such as name, register number, department, and marks can be stored in tables. This type of data can be easily stored, searched and analyzed.

13 Customer comments on social are generally analysis. what type of data is this?

Customer comments on social media are generally considered unstructured data. They are usually written in free-form text and do not follow a fixed table structure.
The comments may contain different words, opinions, symbols, and expressions.
Data science techniques can be used to analyze such text and identify customer opinions.

A company collects a very large amount of daily user activity data from its mobile app. which Big Data characteristic is applied?

The Big Data characteristic involved is volume. Volume refers to the very large amount of data generated and collected. A mobile application can produce huge amount of user activity data every day. Big Data technologies are used to store and process such large quantities of data.

15 A hospital dataset contains missing age values and incorrect phone numbers. which data quality issue is present?

The dataset has data quality issues, particulary missing and inaccurate data. Missing age values mean that some required information is not available.
Incorrect phone numbers represent inaccurate or invalid information in the dataset.
Such issues should be identified and corrected data cleaning.

16 A business wants to convert customer data into useful sales decisions. which Big Data characteristic is important?

The important Big Data characteristic is value, because data should provide useful information for business decisions.
These insights can then be used to make better business and sales decisions.

17 A company stores large files on Hadoop Distributed File System. which part of the Big Data ecosystem is used?
The component used is Hadoop Distributed File System (HDFS)
HDFS is designed to store very large datasets across multiple computers. It divides data into blocks and stores them across the distributed system, provides Big Data applications.

18 A company wants to process large datasets using parallel computation. which Big Data tool can be used?
Apache Spark can be used because it supports fast, distributed and parallel processing of large datasets. It can process large amount of data efficiently and supports different data processing Big tasks.

19 An Organization wants to collect live sensor data from IOT devices. which Big Data ecosystem component is suitable.
A streaming/data ingestion component such as Apache. Kafka can be used to collect and handle live sensor data.
The collected data can then be processed and analyzed using Big Data tools.

20 A company wants to visualize Big Data results for decision-makers. which tool or method can be applied?
Data visualization tools such as dashboards and charts can be used to present Big Data results clearly to decision-makers. This helps organizations make informed decisions based on data.